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RESEARCH NOTE

A palaeomagnetic study of Cainozoic basalts from Northern Nigeria

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SUMMARY

Palaeomagnetic results are presented for 95 specimens from 10 sites from the Jos Plateau Cainozoic basalts in the Northern Nigeria volcanic province. Available K–Ar age determinations from this area suggest that the rocks formed during the Pleistocene. After alternating field demagnetization, the mean direction of the natural remanent magnetization is $D = 341.7°$, $I = -8.7°$ with $\alpha_{95} = 4.2°$. This yields a virtual geomagnetic pole position of 67.0°S, 63.0°E.

An estimated average palaeomagnetic latitude of about 4.4°S for the area of study when compared with the present geographic latitude of about 9.5°N, implies a latitudinal northward drift of about 14° for the African plate since the time of formation of the rocks. This is incompatible with current models for plate motions and most likely represents the effects of conditional local tectonic movements or departures of the geomagnetic field from an axial dipole configuration.

Key words: Jos Plateau Cainozoic basalts, palaeomagnetism.

INTRODUCTION

Volcanic activity has occurred intermittently from Tertiary to very recent times in Northern Nigeria, the products of which are mainly basaltic. This study is aimed at presenting a palaeomagnetic analysis of the Tertiary basalts from the Jos Plateau, an area with relatively young volcanic rocks in Northern Nigeria. Marton & Marton (1976) reported palaeomagnetic pole positions from two separate Pleistocene basalt flows from Biu and Longuda Plateau (between latitudes 10.3°N and 10.7°N and longitudes 11.8°E and 12.9°E). This set of data generally exhibited a large dispersion of the directions of total natural remanent magnetization (NRM) but this was reduced after alternating field demagnetization. The remanent magnetism of the Vom Volcanics (latitude 9.7°N and longitude 8.9°E) were studied by Nwachuku (1977). He reported a stable site mean direction of declination = 354° and inclination = −6° from measurements on six cores drilled at this site, which has been dated as 0.9 ± 0.2 Ma by the potassium–argon method (Grant, Rex & Freeth 1972). Since the volcanicity in this area spans a long interval of time during which the earth's main geomagnetic field has undergone several polarity reversals, Nwachuku (1977) suggested a 'piecemeal' approach with appropriate good age control for meaningful interpretation as a suitable approach to the palaeomagnetic study of these volcanics.

The present study is spread over 10 different sites on and around the Jos Plateau between latitudes 9.0° and 10.0°N and longitudes 8.5° and 9.5°E.

GEOLOGY

The igneous rocks of post-Palaeozoic age in Nigeria can be divided into the following three main groups; the Jurassic Younger Granites and associated rocks, the Cretaceous intrusions and the products of Tertiary to Recent volcanic activities.

Two different periods of volcanic activity in the Cainozoic era have been revealed by K–Ar age determinations: the earlier is phonolitic–trachytic and is dated at 22–11 Ma while the later, more widespread and chiefly basaltic, activity began about 7 Ma and has continued until recent times (Grant *et al.* 1972). The Jos Plateau, together with the highland plains of Hausaland in Northern Nigeria, form the North Central Plateau of Nigeria. The volcanic activity which has taken place intermittently from Tertiary to very recent times in this area gave rise to products which are almost entirely basaltic. Turner (1971), subdivided these products into three groups: lateritized older basalts, older basalts and the newer basalts.

The newer basalts occupy nearly 380 km^2 in the western and southern plateau and cover a considerable area below the escarpments in the west and southeast. They were

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erupted after the plateau had attained almost its present-day topography and are themselves little affected by erosion. The volcanoes generally rise only a few hundred metres above the plateau surface and are steepsided with a central crater, which in some places measures more than half a kilometre in diameter. Some of the cones are built up of ash and scoriaceous lava while others are of more massive basalt. The thickness of the basalt flows is limited by the depth of the pre-existing valleys and rarely exceeds about 300 m in most cases, while the more extensive basalt sheets are made up of several flows. The large Kassa newer basalt (Fig. 1) includes several distinct petrographic types, and drilling to the west of Kassa has intercepted a horizon of sediments and decomposed rock between two distinct basalt flows. The basalts of the western plateau and the Panyam volcanic line have been described by Falconer (1926) and Mackay, Greenwood & Rockingham (1949).

A group of large lateritized basalt remnants lie to the south of Jos on each side of the Bukuru road in the vicinity of the old airport. Lateritized basalts are also well exposed in the Werram, Kwi, Kuru, Keffi Abo and Mongona areas (Fig. 1). The older basalts are generally small, eroded and partly decomposed remnants without a well-defined volcanic focus. Numerous small basalts outcrops in the Werram valley, Rukuba, Ganawuri, South Rop and Monga areas belong to this group (Fig. 1). Many of the older basalts of the Pankshin, Mongun and Mushere outcrops are considered to be later than the lateritization and so rest upon gently sloping erosion surfaces cut into the laterite.

The younger Cainozoic volcanic activity which affected parts of the Jos Plateau led to the formation of the volcanic rocks, often referred to as the Newer Basalts, in that area with an age range of 0.7–22 Ma (Macleod, Turner & Wright 1971). Grant *et al.* (1972), using the K–Ar method, dated rocks from Kass Hill as 1.5 ± 0.1 Ma, Vom as 0.9 ± 0.2 Ma and Bassa as 2.1 ± 0.1 Ma. All these localities are from the rock unit A of this study which consists of sites 1 to 8. The ages of 1.5 and 2.1 for Kass Hill and Bassa respectively from the same group of rocks show that the volcanicity spanned a significant interval of time during which the earth's field may have undergone some polarity reversals. The samples from rock unit B (sites 9 and 10) show a reversed polarity. More K–Ar age determinations for rocks in this area are still to be carried out.

SAMPLING PROCEDURE

The newer basalt rocks of the Jos Plateau may be grouped into two areas: the 'Western Group' (rock unit A) which is the major area lying between latitudes 9.4° and 10.0°N and longitudes 8.5° and 9.0°E and the 'Southeastern Group' (rock unit B), a relatively smaller group lying between

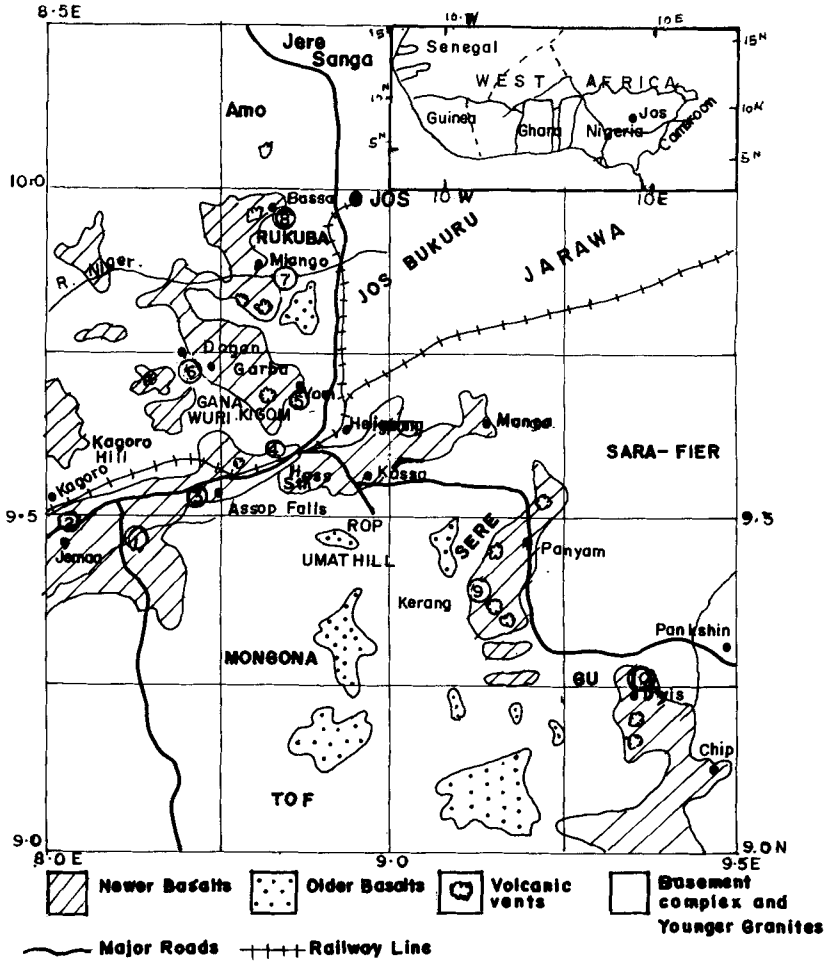


Figure 1. Synoptic map showing distribution of basalt rocks of the Jos Plateau and the sampling sites.

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Table 1. Sampling site statistics.

Locality	Site Code	No. of Samples	No. of Specimens	Site positions Latitude (N)	Longitude (E)
(1) Gimi (River)	GR	8	10	9.48°	8.52°
(2) Kagoro	KA	9	10	9.52°	8.50°
(3) Assop Falls	AF	8	9	9.58°	8.67°
(4) Hoss Station	HS	9	12	9.63°	8.70°
(5) Vom (river)	VM	9	11	9.70°	8.87°
(6) Dogon Garba	DG	8	10	9.72°	8.73°
(7) Miango	MG	8	8	9.78°	8.83°
(8) Bassa	BA	9	9	9.87°	8.87°
(9) Kerang	KG	8	8	9.25°	9.13°
(10) Dyis	DY	8	8	9.17°	9.35°

The locations of the sampling sites are shown on the map in Fig. 1.

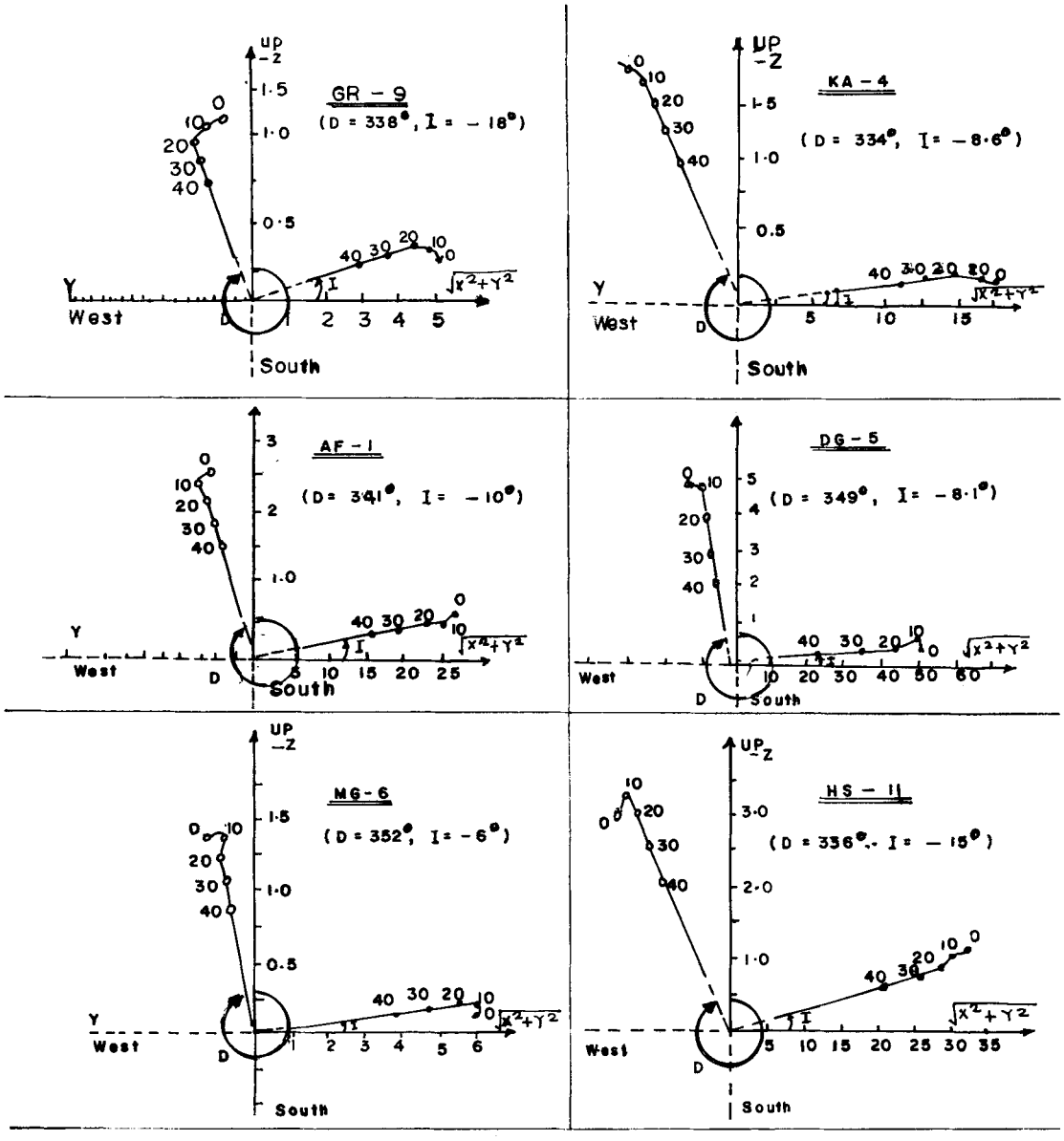


Figure 2. Vector diagrams illustrating the changes in direction and intensity (in A m^{-1}) of remanence during progressive demagnetization of some pilot samples. Solid and open symbols represent inclination and declination respectively. The demagnetization steps were 0, 10, 20, 30, and 40 mT peak value alternating field. Units of both axes are standardized to 1×10^{-1} A m^{-1}.

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latitudes 9.0° and 9.5°N and longitudes 9.2° and 9.4°E (Fig. 1). Sampling covered more than three-quarters of the total area of these two major groups within the newer basalt rocks.

Generally a minimum of eight oriented samples was collected from each of the 10 sampling sites shown in Fig. 1 on and around the Jos Plateau. Hand samples were collected mainly from road cuts, quarries, river beds or stream sections, as well as some natural surfaces not badly weathered and least likely to be affected by lightning. Great care was taken to select fresh material to avoid the effects of alteration. Orientation of the samples was made by means of a magnetic compass with orientation errors estimated to be about ±2°. The sampling site statistics are given in Table 1.

From the oriented blocks, cores 2.0 cm in diameter were drilled in the laboratory. From each of these cores one or more cylinders, 2.5 cm in length, were cut. For the purpose of statistical analysis, the mean of all the cylinders from a single block was regarded as a single observation. A total of 95 samples from the 10 different localities were used in this study while the number of specimens per sample varied between 1 and 3. Further errors involved during reorienting the samples for drilling and during the transference of orientation marks to the individual cores may make the total accuracy of orientation probably (±3 to 5)° and may be systematic if caused by incorrect reorientation.

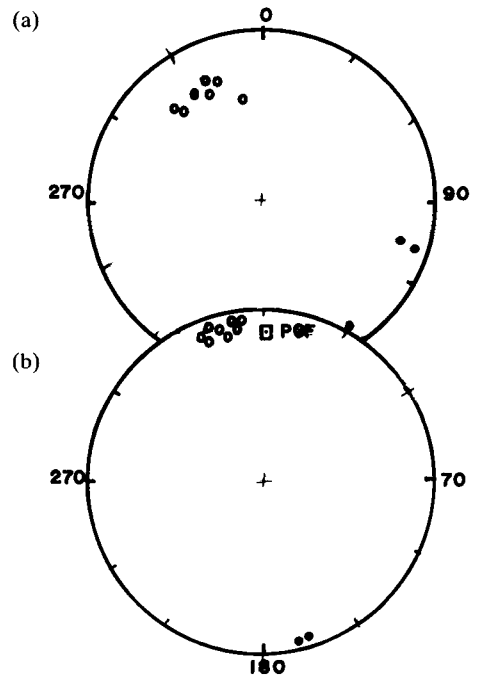


Figure 3. Site mean palaeomagnetic directions (a) before and (b) after demagnetization (closed and open symbols indicate positive and negative inclinations respectively). PGF = present geomagnetic field.

PALAEOMAGNETISM

The direction and intensity of the natural remanent magnetization (NRM) was measured using an astatic magnetometer. The required field free space was provided by a system of square Helmholtz coils. The mean direction of remanent magnetization and statistical parameters for each site were obtained using the method of Fisher (1953).

Alternating field (AF) demagnetization was used to test the stability of NRM of each group of samples. The AF demagnetization was carried out using a rotatory tumbling device for simultaneously rotating the specimen about two mutually perpendicular axes while the demagnetizing alternating field was being applied.

The removal of low-stability magnetization and hence reduction of scatter is done by progressive demagnetization of the specimens in alternating fields until good stability is achieved (usually with an applied field of about 30 mT in this study). Pilot specimens were selected from different sites and used to determine the adequate required field for demagnetization. All other specimens were then subjected to demagnetization in alternating fields of about 30 mT. The Zijderveld diagrams for determining the endpoint of demagnetization for some selected samples are shown in Fig. 2.

Figure 3 shows the site mean magnetization directions on a lower hemisphere equal-area projection. The directions of

Table 2. Site mean directions of magnetization.

		NRM							Demagnetized							
Site	*N*	*M*	*D*	*I*	*R*	*K*	α_{95}	*Hac*	*M*	*D*	*I*	*R*	*K*	α_{95}	λ	*Q*	
+GR	10	18	346.2	−47.3	3.3	1.4	38.1	30	16	342.6	−13.9	9.8	45.0	6.6	7.1S	3.4
+KA	10	88	327.1	−18.9	8.0	4.4	21.1	30	82	338.1	−9.6	9.9	60.0	5.7	4.8S	4.2
+AF	9	89	331.9	−15.2	2.7	2.4	30.1	35	78	332.2	−11.0	8.8	40.0	7.0	5.6S	4.4
HS	12	26	311.2	−21.0	2.8	1.2	36.1	30	24	326.1	−8.2	11.9	110.0	3.9	4.1S	4.3
VM	11	25	336.4	−19.3	5.2	1.7	32.0	30	21	349.0	−7.4	10.9	100.0	4.2	3.7S	2.1
DG	10	145	312.8	−16.8	2.3	1.2	41.2	30	126	348.9	−8.4	9.9	225.3	3.0	4.2S	2.1
MG	8	173	327.3	−19.9	2.6	1.3	42.4	30	138	349.8	−6.6	8.0	301.7	2.9	3.3S	2.1
BA	9	46	106.9	4.1	4.9	1.9	34.1	35	38	338.7	−7.1	8.9	133.0	4.1	3.6S	3.4
+KG	8	96	107.3	15.0	4.2	1.8	33.8	30	74	164.9	7.6	7.9	280.0	3.1	3.8N	2.1
DY	8	88	154.5	14.9	7.0	6.7	19.2	30	75	165.4	7.1	8.0	233.3	3.2	3.6N	2.2

N: number of specimens, *M*: the mean intensity in A $m^{-1} \times 10^{-2}$, *D*: declination (east of north), *I*: inclination (positive downwards), *R*: resultant of the *N* unit vectors, *K*: Fisher dispersion parameter, α_{95}: Semi-angle of 95 per cent cone of confidence, λ: palaeomagnetic latitude, *Hac*: peak AF used (in mT), *Q*: the Koenigsberger ratio, +: one sample with antiparallel magnetization inverted in calculating the site mean statistics after cleaning.

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NRM and that of the characteristic remanent magnetization (ChRM) after 'cleaning' by alternating field demagnetizing are shown for easy comparison. From Table 2 it can be observed that the NRM directions were generally widely scattered, but after cleaning, the characteristic remanent magnetization (ChRM) vectors became clustered together showing a consistent stable direction of magnetization.

MEAN DIRECTIONS AND POLE POSITIONS

Table 2 gives the site mean directions of magnetization before and after alternating field demagnetization, which are plotted on an equal-area projection in Fig. 3.

The virtual geomagnetic pole calculated for each of the ten sites is listed in Table 3, while Fig. 4 shows polar equal-area projection of these poles. Sites 1 to 8 are normally magnetized whereas sites 9 and 10 are of reverse polarity.

INTENSITY OF MAGNETIZATION

The magnetic intensities of the samples are shown in the histogram of Fig. 5 while the mean value of intensity for each site is given in Table 2.

Table 3. Palaeomagnetic poles calculated from 10 site mean directions for the Jos Plateau basalt rock complex.

	Site	Lat.	Pole Long.	δp	δm
(1)	GR	66.3°S	54.7°E	3.5°	6.7°
(2)	KA	63.9°S	66.2°E	2.9°	5.8°
(3)	AF	57.6°S	72.0°E	3.4°	7.1°
(4)	HS	53.5°S	78.1°E	2.0°	3.9°
(5)	VM	72.6°S	48.0°E	2.1°	4.2°
(6)	DG	72.3°S	47.3°E	1.5°	3.0°
(7)	MG	73.4°S	47.0°E	1.5°	2.9°
(8)	BA	64.9°S	67.5°E	2.0°	3.9°
(9)	KG	70.5°N	140.4°E	1.6°	3.1°
(10)	DY	70.6°N	140.1°E	1.6°	3.2°
	Mean 1–8	66.2°S	62.7°E	2.7°	5.3°
	Mean 9–10	70.4°N	139.9°E	—	—
	Mean 1–10*	67.0°S	63.0°E	2.2°	4.2°

δp and δm define the oval of 95 per cent confidence about the pole position. (δp is the error that lies along the great circle passing through the point of observation, S, and the palaeomagnetic pole, P, and is the error in determining the distance between P and S. δm is the error in declination which corresponds to displacement δm from P in the direction perpendicular to the great circle PS.)

* After inversion of poles from sites 9 and 10.

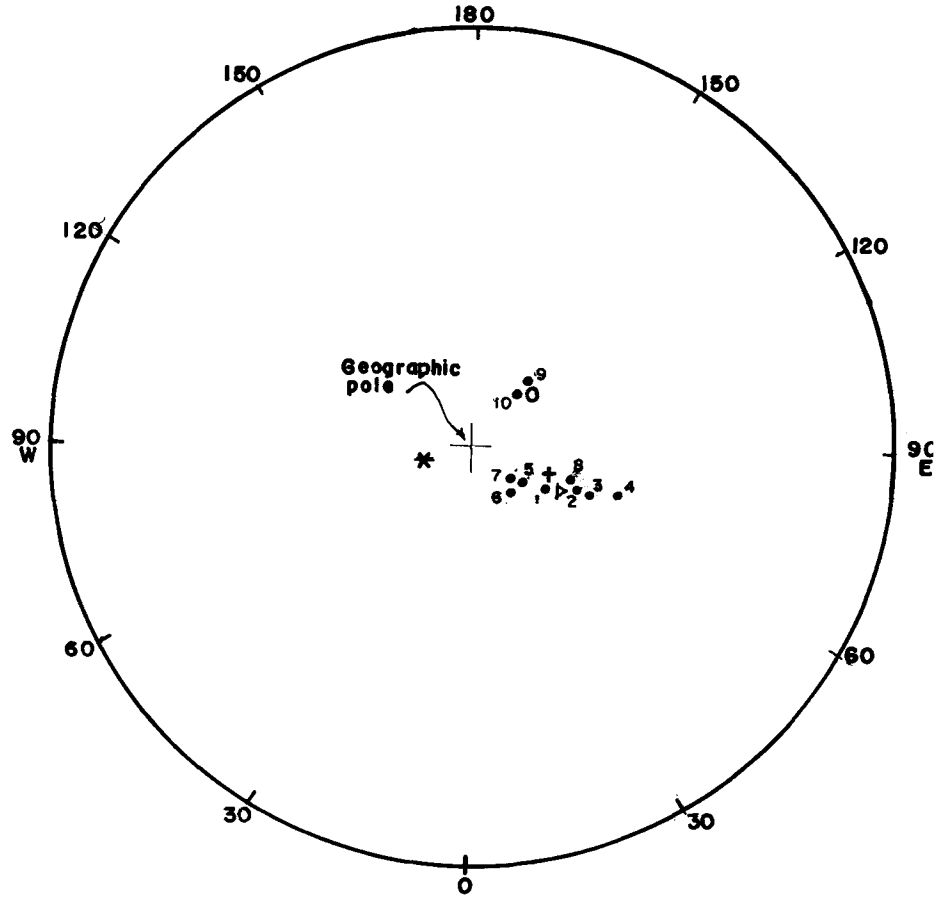


Figure 4. Palaeomagnetic pole: poles are plotted on an equatorial equal-area projection. ● Sites 1–10 pole position, ▷ mean for sites 1–8, ○ mean for sites 9 and 10, + mean for sites 1–10, * present pole position.

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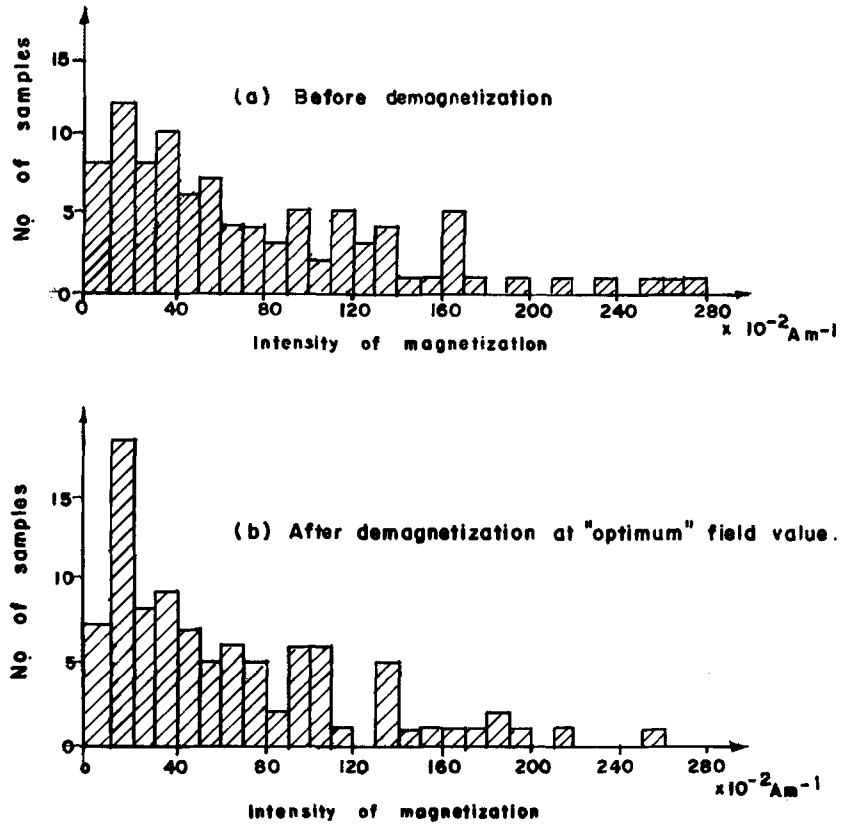


Figure 5. Histograms of the intensity of magnetization before and after AF demagnetization.

The mean value of intensity of remanence varies over a wide range of about 0.25 to 1.73 A m^{-1} before demagnetization and 0.21 to 1.38 A m^{-1} after demagnetization (Table 2); however, the overall variation from all the samples is 0.05 to 2.80 A m^{-1}. This may reflect many factors such as variations in the percentage of magnetic minerals in the rock, intensity of the geomagnetic field at the time of magnetization and the degree to which the original magnetization has changed since it was first acquired.

The distributions of magnetic intensity values before and after demagnetization are similar, with peaks at low values and tailing off towards high values. Both histograms have approximately lognormal shapes.

The variation in remanence intensity is unlikely to be directly related to varying grain size as this would be reflected in varying AF characteristics. A more likely explanation could be the variation of the concentration of magnetic minerals between samples.

DISCUSSION

A general directional stability is demonstrated by close clustering of mean site remanence directions after AF demagnetization. The directions obtained for the separately oriented cores from the same site or rock unit are also in close agreement, the within-lava α_{95} value being about 3° to 4° for seven sites (HS, VM, DG, MG, BA, DG and DY). Sites GR, KA and AF have within-lava α_{95} values of 7.4°, 6.4° and 5.9° respectively.

The NRM of most samples proved to be very resistive to demagnetization. During demagnetization in fields up to 30 mT, the magnetic intensities decrease only by about 15 per cent on the average. From the high Q values (Table 2) and the hard resistance to AF demagnetization it may be concluded that the magnetic carriers are predominantly single domain and the basalts are probably highly oxidized.

The major part of the differences in mean directions between sites could be associated with real variations in the local geomagnetic field between the formation of the different units.

The mean position of the virtual geomagnetic pole (VGP) defined from rock unit A (sites 1–8) is found to be 66.2°S, 62.7°E with $\delta m = 5.3°$ and $\delta p = 2.7°$ relative to the sampling region Jos. The VGP for rock unit B is at 70.4°N, 139.9°E. To estimate the overall VGP for both rock units, the VGPs for rock unit B were first converted to S poles before combining with those from the rock unit A. The overall virtual geomagnetic pole is about 67.0°S, 63.0°E with $\delta m = 4.2°$ and $\delta p = 2.2°$.

The average estimated value of palaeomagnetic latitude ($\lambda_0 = 4.4°$S) compared with the present average latitude of sampling sites ($\lambda_s = 9.5°$N) suggests a possible northwards drift of the African plate since formation of those rock units. Although it has long been suggested that the African plate has made an appreciable northwards drift since the beginning of the break up of the Pangaea some 225 Myr ago (Dietz & Holden 1970), a drift of about 14° is considerably higher than the value expected for rocks of about 1–2 Ma.

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CONCLUSIONS

The Cainozoic basalts from the Jos Plateau can be subdivided into two groups on the basis of palaeomagnetic results, that is the western group and the southeastern group.

The overall mean direction of the characteristic natural remanent magnetization for the rocks has been estimated to have declination $D = 341.7°$ and inclination $I = -8.7°$ with $\alpha_{95} = 4.2°$. This has yielded a virtual geomagnetic pole at 67.0°S, 63.0°E with $\delta p = 2.2°$ and $\delta m = 4.2°$.

From the available radiometric age data (Grant *et al.* 1972), the age of these basalt rocks can be put as about 1.5 ± 0.3 Ma. This falls in the Pleistocene geological epoch of the Quaternary period, with the normal polarity correlated with the Olduvia subchron of the Matuyama reversed magnetic polarity chron. The presence of two different polarities in one rock complex may suggest that the geomagnetic field reversed during emplacement of this complex. A time span of at least 10^4 yr should thus likely separate consecutive volcanic pulses at the localities of rock unit A (northwestern region) from those of rock unit B (southeastern region). This is based on the fact that for opposite magnetization polarities, the minimum age difference is only of the order of 10^4 yr, the time required for the earth's magnetic field to complete a reversal (Fuller, Williams & Hoffman 1979). One interesting observation is that although rocks of both polarities are observed, their directions are fairly antiparallel, and both are a long way from the geocentric axial dipole field direction. From this, it can be concluded that they must all have formed over a short time interval during which there was a field reversal and possibly also an appreciable consistent non-dipole contribution to the geomagnetic field.

The average palaeomagnetic latitude of 4.4°S compared to the average sampling site latitude of 9.5°N cannot be reasonably explained by northwards drift over the last 1.5 Myr. The discrepancy corresponding to over 2000 km northward displacement and about 19° of anticlockwise rotation over this period is not compatible with the geology of the African continent. A most likely cause for the discrepancy is the local geological structures of the sampling area. There was a limit to the degree of accuracy of the orientation marks (i.e. strike and dip), especially for samples collected from the conical hills which have been ignored. Thus orientation errors could also contribute to this discrepancy. However they are unlikely to be systematic.

Furthermore, it is unlikely that a reasonable time-average pole could be deduced from only 10 sites in a volcanic sequence, hence secular variation may not be averaged out. However the apparent presence of a polarity reversal record in these lavas does suggest that their emplacement extended over a time interval of about 10^4 yr. It is considered premature at present to make conclusions about the apparent polar wandering.

It is observed that the palaeofield direction distributions in Fig. 3 are highly elongated in the horizontal direction, especially if the distortions of the projection are considered. Such elongation might have arisen from incompletely resolved remanence components in two different directions. The characteristic remanent magnetization could lie at one end of the pattern while in such young underformed rocks there are a limited number of possible origins for a secondary magnetization direction. It is most likely to arise as viscous remanent magnetization (VRM) especially as the samples were stored in a systematic manner.

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